

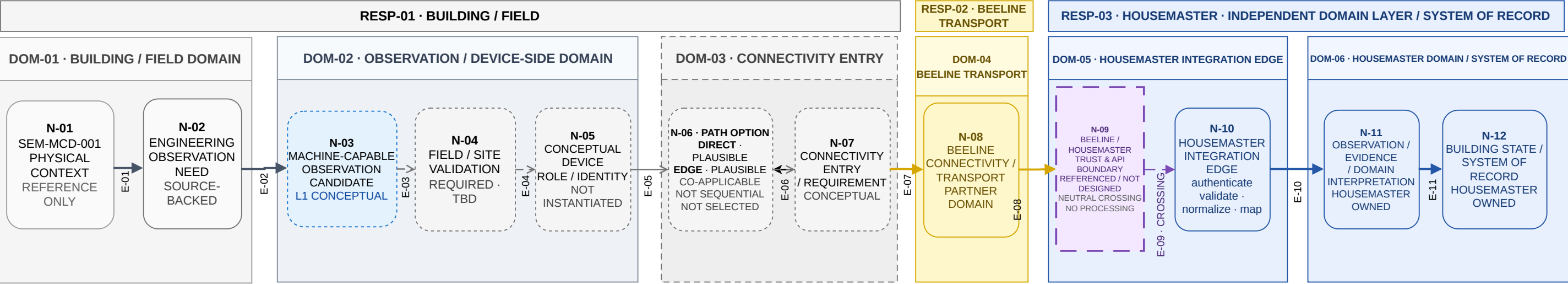
A3-EC03-01

REFERENCE · SEM-MCD-001
Semey, Shakarima 13A
Deployment: NOT ESTABLISHED ·
Installed devices: 0 CONFIRMED

CONNECTIVITY ENTRY ARCHITECTURE

Connectivity serves an observation requirement; it does not define the engineering meaning of the observation.

FROZEN SEMANTICS
DRAWIO REVIEW



THREE ARCHITECTURAL PLANES

DATA PLANE
Readings / states / scalar / event / counter
Preserve original **event_time**

CONTROL PLANE · NOT ENABLED BY DEFAULT
Remote actuation out of scope.

OPERATIONS PLANE
Connectivity status · reachability · lifecycle · diagnostics
Exact Beeline exposure: TBD

MINIMUM INTEGRATION SURFACE · CONCEPTUAL

- 1 · Observation/event content
scalar · event · state · counter
- 2 · Source/device + connectivity-profile identity references
TBD until deployment
- 3 · Connectivity / delivery status
exposed mapping **TBD WITH BEELINE**
- 4 · Temporal provenance · **event_time** ≠ **received_time**
- 5 · Integration outcome
validate · normalize · map · no payload schema
HouseMaster responsibility

OPEN / TBD REGISTER · 20 UNRESOLVED

SITE / DEVICE FEASIBILITY — 5
TBD-EC03-01 · power availability
TBD-EC03-02 · physical access
TBD-EC03-03 · RF/connectivity
environment
TBD-EC03-04 · Direct vs Edge
preference
TBD-EC03-05 · metering-interface
existence

TBD WITH BEELINE — 15
Q-CONN-01 · service models
Q-CONN-02 · SIM/eSIM lifecycle
Q-CONN-03 · exposed states
Q-CONN-04 · delivery semantics
Q-CONN-05 · retry/reconnect ownership
Q-CONN-06 · authentication model
Q-CONN-07 · private connectivity
Q-CONN-08 · RF/site-survey support
Q-CONN-09 · operator observability
Q-CONN-10 · incident exposure
Q-CONN-11 · provisioning interfaces
Q-CONN-12 · lifecycle events
Q-CONN-13 · onboarding role
Q-CONN-14 · edge/gateway ownership
Q-CONN-15 · pilot monitoring

FAILURE / UNKNOWN BOUNDARY · COMPACT STRIP

DEVICE SIDE → LOCAL POWER / OPTIONAL AGGREGATION → ACCESS / RADIO → OPERATOR TRANSPORT → HOUSEMASTER INTEGRATION EDGE → HOUSEMASTER PROCESSING

NO MESSAGE ≠ NORMAL ENGINEERING CONDITION · CONNECTIVITY LOSS ≠ ENGINEERING DEFECT · DELIVERY FAILURE ≠ PHYSICAL FAILURE · UNKNOWN ≠ ABSENT

MANDATORY GUARDRAILS · 11

